LESSON 8.1

Understand Equations and Their Solutions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

REMEMBER

Understand Equations and Their Solutions

Equations & Solutions. Formula: Left Expression = Right Expression An equation is a math sentence with an equal sign (=). To write one, I pick a letter (a variable) for the unknown number and match the words to math symbols.

1. An equation is a mathematical sentence stating two expressions are equal.
2. A solution is a value for the variable that makes the equation true.
3. Check: Substitute the value into the equation and verify that both sides balance

Words: **Equation *(Ecuación)*** **Variable *(Variable)*** **Equal sign *(Signo igual)*** **Expression *(Expresión)*** **Constant *(Constante)***

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Reflect the point (4, 3) across the x-axis. What are the new coordinates?

(A) (3, 4)
(B) (4, −3)
(C) (−4, 3)
(D) (−4, −3)

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Reflect the point (−2, 5) across the y-axis.

A (2, 5)

B (−2, −5)

C (2, −5)

D (5, −2)

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

When you reflect a point across the x-axis, which coordinate changes sign?

A The y-coordinate

B The x-coordinate

C Both of them

D Neither of them

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

The point (−6, −1) is reflected across the y-axis. In which quadrant does the image land?

A Quadrant I

B Quadrant II

C Quadrant IV

D Quadrant III

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A detective is tracking stolen paintings. She knows that the number of paintings recovered plus 14 paintings still missing equals the 37 total stolen. The equation p + 14 = 37 represents this situation. Solve p + 14 = 37.

(A) p = 14
(B) p = 23
(C) p = 37
(D) p = 51

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Detective's Mistake

1. **Clue** Eight fewer clues than a number n leaves 5 clues.
2. **Choose a variable** Let n be the starting number of clues.
3. **Write the equation** 8 − n = 5

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Does 'eight fewer than a number' start with the 8 or with the number n?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

SENTENCE STARTER I put ___ on the left side because ___, and ___ on the right side because ___. The equal sign means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? □ 4 · I can teach it □ 3 · I've got it □ 2 · I need a hint □ 1 · I need help

LESSON 8.1

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Set B · Version B · Second form · core practice

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- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation for a real situation and check whether a given value is a solution by substituting it.

WHY THIS MATTERS Detective Ruiz received a tip: 'A number of stolen gems plus 8 more equals 20 total gems.' She needs to write an equation to represent this clue before she can solve the case. How would you write this situation as an equation?

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A witness statement reads: 'A number c of security cameras plus 6 more equals 19.' Which equation matches?

(A) c + 6 = 19
(B) 6c = 19
(C) c − 6 = 19
(D) c / 6 = 19

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each detective clue to the equation that cracks it.

_____ A number d of door codes plus 4 equals 17.
_____ 6 times the number of tips t equals 42.
_____ A pile of files f, after losing 11, equals 20.
_____ A number of suspects s split into 5 vans equals 8 per van.

(A) d + 4 = 17
(B) 6t = 42
(C) f − 11 = 20
(D) s / 5 = 8

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A detective is tracking stolen paintings. She knows that the number of paintings recovered plus 14 paintings still missing equals the 37 total stolen. The equation p + 14 = 37 represents this situation. Why subtract 14 from BOTH sides?

(A) To keep the equation balanced while isolating p

(B) Because subtraction is easier than addition

(C) Because 14 is the smaller number

(D) To make the answer a whole number

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A detective is tracking stolen paintings. She knows that the number of paintings recovered plus 14 paintings still missing equals the 37 total stolen. The equation p + 14 = 37 represents this situation. Check your answer: does 23 work?

(A) Yes — 23 + 14 = 37

(B) No — 23 + 14 = 47

(C) No — 23 − 14 = 9

(D) You cannot check an equation

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The lab tech says: '5 times the number of evidence bags b equals 35.' Which equation represents this clue?

(A) 5b = 35

(B) b + 5 = 35

(C) b − 5 = 35

(D) b / 5 = 35

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A torn note reads: 'A suspect's case minutes m, after 9 fewer, leaves 24.' Which equation fits?

(A) m − 9 = 24

(B) m + 9 = 24

(C) 9m = 24

(D) m / 9 = 24

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what expression to put on each side of the balance?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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